

Cálculo del rotacional de $\vec{a}_{SRP}$

En primer lugar, recordamos que el objetivo de este archivo es encontrar los valores de $\alpha, \delta \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ para los cuales $rot(\vec{a}_{SRP})$ en los puntos $(-\mu, 0, 0)$, $(-\mu, 1, 1)$ y $(-\mu, 1, 0)$ es nulo.

En segundo lugar, definimos la función del producto escalar (prodesc), las componentes del vector normal (nx, ny y nz) y las componentes de la aceleración debida a la presión de radiación solar (ax, ay y az).

```
In[*]:= prodesc[x_, y_, z_, alpha_, delta_] :=
  ((Sqrt[(x + mu)^2 + y^2]^2 * Cos[alpha] + z^2) * Cos[delta] + z * Sqrt[(x + mu)^2 + y^2] *
    (1 - Cos[alpha]) * Sin[delta]) / Sqrt[(x + mu)^2 + y^2 + z^2]^2

In[*]:= nx[x_, y_, z_, alpha_, delta_] :=
  1 / Sqrt[(x + mu)^2 + y^2 + z^2] * ((x + mu) * Cos[alpha] * Cos[delta] -
    (x + mu) * z / Sqrt[(x + mu)^2 + y^2] * Cos[alpha] * Sin[delta] -
    y * Sin[alpha] * Cos[delta] + y * z / Sqrt[(x + mu)^2 + y^2] * Sin[alpha] * Sin[delta])

In[*]:= ny[x_, y_, z_, alpha_, delta_] :=
  1 / Sqrt[(x + mu)^2 + y^2 + z^2] * (y * Cos[alpha] * Cos[delta] -
    y * z / Sqrt[(x + mu)^2 + y^2] * Cos[alpha] * Sin[delta] + (x + mu) * Sin[alpha] *
    Cos[delta] - (x + mu) * z / Sqrt[(x + mu)^2 + y^2] * Sin[alpha] * Sin[delta])

In[*]:= nz[x_, y_, z_, alpha_, delta_] :=
  1 / Sqrt[(x + mu)^2 + y^2 + z^2] * (z * Cos[delta] + Sqrt[(x + mu)^2 + y^2] * Sin[delta])

In[*]:= ax[x_, y_, z_, alpha_, delta_] :=
  prodesc[x, y, z, alpha, delta]^2 / Sqrt[(x + mu)^2 + y^2 + z^2]^2 *
    nx[x, y, z, alpha, delta]

In[*]:= ay[x_, y_, z_, alpha_, delta_] :=
  prodesc[x, y, z, alpha, delta]^2 / Sqrt[(x + mu)^2 + y^2 + z^2]^2 *
    ny[x, y, z, alpha, delta]

In[*]:= az[x_, y_, z_, alpha_, delta_] :=
  prodesc[x, y, z, alpha, delta]^2 / Sqrt[(x + mu)^2 + y^2 + z^2]^2 *
    nz[x, y, z, alpha, delta]
```

En tercer lugar, calculamos las componentes de $rot(\vec{a}_{SRP})$. Denotamos por $eq1 = \frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y}$,

$$\text{eq2} = \frac{\partial a_y}{\partial z} - \frac{\partial a_z}{\partial y} \quad \text{y} \quad \text{eq3} = \frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x}.$$

In[*]:= **eq1 = Simplify[D[ay[x, y, z, alpha, delta], x] - D[ax[x, y, z, alpha, delta], y]]**
[simplifica] [deriva]

Out[*]=

$$\begin{aligned} & - \left(\left(\sin[\alpha] \left((z^2 + (\mu^2 + 2\mu x + x^2 + y^2) \cos[\alpha]) \cos[\delta] - \right. \right. \right. \\ & \quad \left. \left. \sqrt{\mu^2 + 2\mu x + x^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \right) \right. \\ & \quad \left(\sqrt{\mu^2 + 2\mu x + x^2 + y^2} (z^2 (5\mu^2 + 10\mu x + 5x^2 + 5y^2 - 2z^2) + \right. \\ & \quad (\mu^2 + 2\mu x + x^2 + y^2) (\mu^2 + 2\mu x + x^2 + y^2 - 6z^2) \cos[\alpha]) \cos[\delta]^2 + \\ & \quad z (3\mu^4 + 3x^4 + 6\mu^2 y^2 + 6x^2 y^2 + 3y^4 - 20\mu x z^2 + z^4 - (5\mu^4 + 20\mu^3 x + 5x^4 + 5y^4 + \\ & \quad 20\mu x (x^2 + y^2) - 9x^2 z^2 - 9y^2 z^2 + \mu^2 (30x^2 - 9z^2)) \cos[\alpha]) \cos[\delta] \\ & \quad \sin[\delta] + z \left(-2 \sqrt{\mu^2 + 2\mu x + x^2 + y^2} z (4\mu^2 + 8\mu x + 4x^2 + 4y^2 - 3z^2) \right. \\ & \quad \left. \sin\left[\frac{\alpha}{2}\right]^2 \sin[\delta]^2 + (6\mu^3 x + 6\mu x (x^2 + y^2) - 5(x^2 + y^2) z^2 + \right. \\ & \quad \left. \left. \mu^2 (9x^2 - 5z^2) + (-5\mu^2 y^2 - 5x^2 y^2 + 9\mu x z^2) \cos[\alpha]) \sin[2\delta] \right) \right) \right) \Bigg) / \\ & \left(\sqrt{(\mu + x)^2 + y^2} ((\mu + x)^2 + y^2 + z^2)^{9/2} \right) \end{aligned}$$

```
In[*]:= eq2 = Simplify[D[ay[x, y, z, alpha, delta], z] - D[az[x, y, z, alpha, delta], y]]
           |simplifica |deriva           |deriva
```

```
Out[*]=
```

$$\begin{aligned}
& \left(\left(z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\deltaelta] - \\
& \quad \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\deltaelta] \Big) \\
& \left(2 \left((\mu + x)^2 + y^2 + z^2 \right) (y \cos[\alpha] + (\mu + x) \sin[\alpha]) \left(\sqrt{(\mu + x)^2 + y^2} \cos[\deltaelta] - \right. \right. \\
& \quad \left. \left. z \sin[\deltaelta] \right) \left(2 z \cos[\deltaelta] - \sqrt{(\mu + x)^2 + y^2} (-1 + \cos[\alpha]) \sin[\deltaelta] \right) - \right. \\
& \quad 2 \left((\mu + x)^2 + y^2 + z^2 \right) \left(z \cos[\deltaelta] + \sqrt{(\mu + x)^2 + y^2} \sin[\deltaelta] \right) \\
& \quad \left(2 y \sqrt{(\mu + x)^2 + y^2} \cos[\alpha] \cos[\deltaelta] - y z (-1 + \cos[\alpha]) \sin[\deltaelta] \right) - \\
& \quad y \left((\mu + x)^2 + y^2 + z^2 \right) \sin[\deltaelta] \left(\left(z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\deltaelta] - \\
& \quad \left. \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\deltaelta] \right) - \\
& \quad \left((\mu + x)^2 + y^2 + z^2 \right) (y \cos[\alpha] + (\mu + x) \sin[\alpha]) \sin[\deltaelta] \\
& \quad \left(\left(z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\deltaelta] - \\
& \quad \left. \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\deltaelta] \right) + \\
& \quad 7 y \sqrt{(\mu + x)^2 + y^2} \left(z \cos[\deltaelta] + \sqrt{(\mu + x)^2 + y^2} \sin[\deltaelta] \right) \\
& \quad \left(\left(z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\deltaelta] - \\
& \quad \left. \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\deltaelta] \right) - \\
& \quad 7 z (y \cos[\alpha] + (\mu + x) \sin[\alpha]) \left(\sqrt{(\mu + x)^2 + y^2} \cos[\deltaelta] - z \sin[\deltaelta] \right) \\
& \quad \left(\left(z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\deltaelta] - \\
& \quad \left. \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\deltaelta] \right) \Big) \Big) / \\
& \left(\sqrt{(\mu + x)^2 + y^2} \left((\mu + x)^2 + y^2 + z^2 \right)^{9/2} \right)
\end{aligned}$$

```

In[*]:= eq3 = Simplify[D[ax[x, y, z, alpha, delta], z] - D[az[x, y, z, alpha, delta], x]]
Out[*]=

$$\left( \left( z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \Big) \left( 2 \left( (\mu + x)^2 + y^2 + z^2 \right) \left( (\mu + x) \cos[\alpha] - y \sin[\alpha] \right) \left( \sqrt{(\mu + x)^2 + y^2} \cos[\delta] - z \sin[\delta] \right) \left( 2 z \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} (-1 + \cos[\alpha]) \sin[\delta] \right) - 2 (\mu + x) \left( (\mu + x)^2 + y^2 + z^2 \right) \left( z \cos[\delta] + \sqrt{(\mu + x)^2 + y^2} \sin[\delta] \right) \left( 2 \sqrt{(\mu + x)^2 + y^2} \cos[\alpha] \cos[\delta] - z (-1 + \cos[\alpha]) \sin[\delta] \right) - (\mu + x) \left( (\mu + x)^2 + y^2 + z^2 \right) \sin[\delta] \left( \left( z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \right) - \left( (\mu + x)^2 + y^2 + z^2 \right) \left( (\mu + x) \cos[\alpha] - y \sin[\alpha] \right) \sin[\delta] \left( \left( z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \right) + 7 (\mu + x) \sqrt{(\mu + x)^2 + y^2} \left( z \cos[\delta] + \sqrt{(\mu + x)^2 + y^2} \sin[\delta] \right) \left( \left( z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \right) - 7 z \left( (\mu + x) \cos[\alpha] - y \sin[\alpha] \right) \left( \sqrt{(\mu + x)^2 + y^2} \cos[\delta] - z \sin[\delta] \right) \left( \left( z^2 + (\mu + x)^2 + y^2 \right) \cos[\alpha] \right) \cos[\delta] - \sqrt{(\mu + x)^2 + y^2} z (-1 + \cos[\alpha]) \sin[\delta] \right) \Big) \Big) / \left( \sqrt{(\mu + x)^2 + y^2} \left( (\mu + x)^2 + y^2 + z^2 \right)^{9/2} \right)$$


```

Por último, calculamos el valor del rotacional en cada uno de los puntos mencionados y resolvemos el sistema de ecuaciones para α y δ .

```

In[*]:= g1 = eq1 /. {x -> -mu, y -> 1, z -> 0}
Out[*]=
-Cos[alpha]^2 Cos[delta]^3 Sin[alpha]

In[*]:= g2 = Simplify[eq2 /. {x -> -mu, y -> 1, z -> 0}]
Out[*]=
(4 - 3 Cos[alpha]) Cos[alpha]^2 Cos[delta]^2 Sin[delta]

In[*]:= g3 = Simplify[eq3 /. {x -> -mu, y -> 1, z -> 0}]
Out[*]=
Cos[alpha] (-2 + 3 Cos[alpha]) Cos[delta]^2 Sin[alpha] Sin[delta]

```

In[*]:= **g4 = Simplify**[eq1 /. {x → -mu, y → 1, z → 1}]
|simplifica

Out[*]=

$$\frac{1}{16\sqrt{2}} \left(\sin[\alpha] (\cos[\delta] + \cos[\alpha] \cos[\delta] + \sin[\delta] - \cos[\alpha] \sin[\delta]) \right. \\ \left. (-1 - 2 \cos[2\delta] + \cos[\alpha] (2 + 3 \cos[2\delta] - 2 \sin[2\delta]) + 3 \sin[2\delta]) \right)$$

In[*]:= **g5 = Simplify**[eq2 /. {x → -mu, y → 1, z → 1}]
|simplifica

Out[*]=

$$\frac{1}{16\sqrt{2}} \left(\cos[\delta] + \cos[\alpha] \cos[\delta] + \sin[\delta] - \cos[\alpha] \sin[\delta] \right) \\ \left(7 \cos[\delta]^2 \sin[\alpha]^2 + \sin[\alpha]^2 \sin[\delta]^2 + \right. \\ \left. 2 (3 - 3 \cos[\alpha] + \cos[2\alpha]) \sin[2\delta] \right)$$

In[*]:= **g6 = Simplify**[eq3 /. {x → -mu, y → 1, z → 1}]
|simplifica

Out[*]=

$$\frac{1}{16\sqrt{2}} \left(\sin[\alpha] (\cos[\delta] + \cos[\alpha] \cos[\delta] + \sin[\delta] - \cos[\alpha] \sin[\delta]) \right. \\ \left. (-1 + 6 \cos[\delta] \sin[\delta] + \cos[\alpha] (4 + 3 \cos[2\delta] - 4 \sin[2\delta])) \right)$$

In[*]:= **g7 = Simplify**[eq1 /. {x → -mu, y → 2, z → 1}]
|simplifica

Out[*]=

$$-\frac{1}{1250\sqrt{5}} \sin[\alpha] \\ \left((1 + 4 \cos[\alpha]) \cos[\delta] - 2 (-1 + \cos[\alpha]) \sin[\delta] \right) (5 + 31 \cos[2\delta] + \\ 9 \cos[\delta] \sin[\delta] + \cos[\alpha] (5 - 21 \cos[2\delta] - 22 \sin[2\delta]))$$

In[*]:= **g8 = Simplify**[eq2 /. {x → -mu, y → 2, z → 1}]
|simplifica

Out[*]=

$$-\frac{1}{625\sqrt{5}} \left((1 + 4 \cos[\alpha]) \cos[\delta] - 2 (-1 + \cos[\alpha]) \sin[\delta] \right)^2 \\ (14 (-1 + \cos[\alpha]) \cos[\delta] + (-13 + 8 \cos[\alpha]) \sin[\delta])$$

In[*]:= **g9 = Simplify**[eq3 /. {x → -mu, y → 2, z → 1}]
|simplifica

Out[*]=

$$\frac{1}{625\sqrt{5}} 2 \sin[\alpha] \left((1 + 4 \cos[\alpha]) \cos[\delta] - 2 (-1 + \cos[\alpha]) \sin[\delta] \right) \\ \left((-13 + 28 \cos[\alpha]) \cos[\delta]^2 + 3 \cos[\delta] \sin[\delta] - \right. \\ \left. 8 (-1 + \cos[\alpha]) \sin[\delta]^2 + \cos[\alpha] \sin[2\delta] \right)$$

```

In[*]:= Reduce[{g1 == 0, g2 == 0, g3 == 0, g4 == 0, g5 == 0, g6 == 0, g7 == 0, g8 == 0,
|reduce
      g9 == 0, -Pi / 2 ≤ alpha ≤ Pi / 2, -Pi / 2 ≤ delta ≤ Pi / 2}, {alpha, delta}]
      |número pi      |número pi |número pi      |número pi
Out[*]=
alpha == 0 && (delta == 0 || delta == - $\frac{\pi}{2}$  || delta ==  $\frac{\pi}{2}$ )

```